



IoT Security Analyst

QP Code: SSC/Q8207

Version: 4.0

NSQF Level: 6

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Qualification Pack

Contents

SSC/Q8207: IoT Security Analyst	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
SSC/N8231: Design processes to manage security threats and incidents across multiple technology layers	5
SSC/N8232: Detect occurrences of security incidents to IoT solutions	9
SSC/N8233: Respond to detected security incidents and restore affected capabilities	14
SSC/N8238: Create technical documents and manuals	19
SSC/N8249: Operational Technology Security	23
DGT/VSQ/N0102: Employability Skills (60 Hours)	27
Assessment Guidelines and Weightage	34
<i>Assessment Guidelines</i>	34
<i>Assessment Weightage</i>	34
Acronyms	36
Glossary	37

Qualification Pack

SSC/Q8207: IoT Security Analyst

Brief Job Description

An IoT Security Analyst is a professional specializing in protecting Internet of Things (IoT) devices, networks, and systems from threats and vulnerabilities. They design, implement, manage, and monitor security measures tailored to IoT environments, ensuring the confidentiality, integrity, and availability of data and operations. Their role involves understanding the unique security challenges of IoT devices such as limited processing power, diverse protocols, and widespread deployment and applying best practices, standards, and open-source tools to safeguard these assets. They also conduct vulnerability assessments, develop incident response plans, and ensure compliance with relevant security standards. Overall, an IoT Security Specialist helps organizations maintain resilient and secure IoT ecosystems critical for modern businesses and infrastructure.

Personal Attributes

IoT Security Specialists must have good interpersonal and communication skills. They must have a zeal to develop their knowledge and analytical abilities.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8231: Design processes to manage security threats and incidents across multiple technology layers](#)
2. [SSC/N8232: Detect occurrences of security incidents to IoT solutions](#)
3. [SSC/N8233: Respond to detected security incidents and restore affected capabilities](#)
4. [SSC/N8238: Create technical documents and manuals](#)
5. [SSC/N8249: Operational Technology Security](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things

Qualification Pack

Country	India
NSQF Level	6
Credits	19
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2522.0201
Minimum Educational Qualification & Experience	Completed 4 year UG program (PHD or PG or UG or diploma with courses related to Engg./ Science) OR Completed 3 year UG degree (PHD or PG or UG or diploma with courses related to Engg./ Science) with 1.5 years of experience in relevant field.Relevant Experience in job roles related to Computer Science/IT/Network Security/Cybersecurity/IoT The relevant experience would include work, internship, and apprenticeship after completing relevant educational qualifications. OR Previous relevant Qualification of NSQF Level (5.5) with 1.5 years of experience in relevant field.Relevant Experience in job roles related to Computer Science/IT/Network Security/Cybersecurity/IoT The relevant experience would include work, internship, and apprenticeship after completing relevant educational qualifications
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	22 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	4.0
Reference code on NQR	QG-06-IT-045982025-V2-NASSCOM
NQR Version	2

Qualification Pack

SSC/N8231: Design processes to manage security threats and incidents across multiple technology layers

Description

This unit is about designing processes to mitigate security risk at the network, hardware, cloud, application and platform layers

Scope

The scope covers the following :

- Manage access to IoT assets, Ensure data security, Ensure network and hardware security, Design network security, Define security architecture, Ensure cloud, application and platform security, Define fallback mechanisms

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Manage access to IoT assets and systems based on identity access and management tools and frameworks
- PC2.** Evaluate security threat levels and criticality of IoT assets and systems to prioritize security measures
- PC3.** Define security parameters and develop encryption standards for data stored in cloud platforms and databases
- PC4.** Maintain logs of data security performance metrics and analyze variances to identify issues and take action
- PC5.** Conduct security assessments including penetration testing and vulnerability scans using automated and manual techniques
- PC6.** Protect the IoT network and devices from unauthorized access by configuring and troubleshooting security hardware and software
- PC7.** Secure the perimeter of the IoT network and classify components based on function and data sensitivity
- PC8.** Employ network monitoring tools and design logging systems to detect and respond to security threats in real time
- PC9.** Ensure all security policies and standards are consistently followed and security processes are managed daily
- PC10.** Regularly apply security patches and employ digital certificates along with disaster recovery plans to ensure resilience

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to maintaining solution security

Qualification Pack

- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies for documenting and implementing security procedures
- KU4.** who to involve while defining and executing security processes
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** the fundamentals of security and the various levels that need to be covered such as device, cloud, communications, applications, etc.
- KU7.** how to communicate security requirements and processes across all levels of the organization
- KU8.** how to create and update Intrusion Detection Systems (IDS)
- KU9.** how to create and update Security Information Event Management (SIEM) Tools
- KU10.** how to define processes for handling security across various solution layers
- KU11.** how to manage Malware Software
- KU12.** how to manage Intrusion Prevention Systems
- KU13.** how to install firewalls and data encryption protocols
- KU14.** the internal and external security regulations and standards
- KU15.** how to prepare and submit security assessment reports and whom to share them with
- KU16.** how to keep track of changes, procedures, and practices
- KU17.** how to store and retrieve information

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** apply problem-solving approaches to different situations
- GS7.** analyze the business impact and disseminate relevant information to others
- GS8.** apply balanced judgments to different situations
- GS9.** check the work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Manage access to IoT assets and systems based on identity access and management tools and frameworks	3	5	-	2
PC2. Evaluate security threat levels and criticality of IoT assets and systems to prioritize security measures	3	5	-	2
PC3. Define security parameters and develop encryption standards for data stored in cloud platforms and databases	3	5	-	2
PC4. Maintain logs of data security performance metrics and analyze variances to identify issues and take action	3	5	-	2
PC5. Conduct security assessments including penetration testing and vulnerability scans using automated and manual techniques	3	5	-	2
PC6. Protect the IoT network and devices from unauthorized access by configuring and troubleshooting security hardware and software	3	5	-	2
PC7. Secure the perimeter of the IoT network and classify components based on function and data sensitivity	3	5	-	2
PC8. Employ network monitoring tools and design logging systems to detect and respond to security threats in real time	3	5	-	2
PC9. Ensure all security policies and standards are consistently followed and security processes are managed daily	3	5	-	2
PC10. Regularly apply security patches and employ digital certificates along with disaster recovery plans to ensure resilience	3	5	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8231
NOS Name	Design processes to manage security threats and incidents across multiple technology layers
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	6
Credits	5
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8232: Detect occurrences of security incidents to IoT solutions

Description

This unit is about detecting threats and anomalies, employing continuous monitoring processes, maintaining detection processes and performing audits

Scope

The scope covers the following :

- Review past incidents, Drive audit operations, Perform security analysis, Drive continuous monitoring and threat assessment

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** record, classify and prioritize information security incidents using standard templates and tools
- PC2.** access their organization's knowledge base for information on previous information security incidents and how these were managed
- PC3.** participate in audit reviews
- PC4.** identify the requirements of audits and prepare for audits in advance
- PC5.** liaise with appropriate people to gather data/information required for audits
- PC6.** provide immediate support to auditors to carry out audit tasks
- PC7.** carry out required audit tasks using standard tools and following established procedures/guidelines/checklists
- PC8.** evaluate the impact of potential anomalies
- PC9.** ensure both signature-based and anomaly-based intrusions are detected in a timely manner
- PC10.** perform regular maintenance of detection processes
- PC11.** ensure readiness of maintenance processes to reliably detect incidents
- PC12.** apply security controls to IoT solutions in line with data security policies, procedures and guidelines
- PC13.** perform security assessments of gateway / edge devices systems using automated security assessment tools
- PC14.** design and develop dashboards/KPIs for monitoring the incidents, the root cause and detection to resolution corrective actions using standard Commercial off-the-shelf based Investigation Management tool
- PC15.** identify critical security threats and vulnerabilities across all layers of the IoT solution
- PC16.** predict and extrapolate attack trends ahead of their occurrence
- PC17.** carry out backups of security devices and applications in line with security policies, procedures and guidelines

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to maintaining solution security
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies for documenting and implementing security procedures
- KU4.** who to involve while defining and executing security processes
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** the fundamentals of security and the various levels that need to be covered such as device, cloud, communications, applications, etc.
- KU7.** the routine operational procedures and tasks required to be coordinated and responded to
- KU8.** how to identify and resolve security vulnerabilities and incidents
- KU9.** the common security issues and incidents that may require action and whom to report them to
- KU10.** how to maintain records of monitoring activities
- KU11.** how to obtain and validate information related to security issues
- KU12.** how to prepare and submit security reports and whom to share them with
- KU13.** how to query and write scripts
- KU14.** how to identify and refer anomalies in data
- KU15.** how to store and retrieve information
- KU16.** how to keep track of changes, procedures, and practices

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** ask for clarification and advice from appropriate people
- GS3.** listen effectively and communicate information accurately
- GS4.** follow rule-based decision-making processes
- GS5.** make decisions on suitable courses
- GS6.** plan and organize the work to achieve targets and meet deadlines
- GS7.** refer anomalies to the supervisor
- GS8.** analyze the business impact and disseminate relevant information to others
- GS9.** apply balanced judgments to different situations
- GS10.** check the work is complete and free from errors
- GS11.** work independently and collaboratively

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. record, classify and prioritize information security incidents using standard templates and tools	2	2	-	1
PC2. access their organization's knowledge base for information on previous information security incidents and how these were managed	2	2	-	1
PC3. participate in audit reviews	2	3	-	1
PC4. identify the requirements of audits and prepare for audits in advance	3	4	-	2
PC5. liaise with appropriate people to gather data/information required for audits	1	3	-	1
PC6. provide immediate support to auditors to carry out audit tasks	1	3	-	1
PC7. carry out required audit tasks using standard tools and following established procedures/guidelines/checklists	1	3	-	1
PC8. evaluate the impact of potential anomalies	2	2	-	1
PC9. ensure both signature-based and anomaly-based intrusions are detected in a timely manner	2	2	-	1
PC10. perform regular maintenance of detection processes	1	3	-	1
PC11. ensure readiness of maintenance processes to reliable detect incidents	3	4	-	2
PC12. apply security controls to IoT solutions in line with data security policies, procedures and guidelines	1	3	-	1
PC13. perform security assessments of gateway / edge devices systems using automated security assessment tools	3	6	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. design and develop dashboards/KPIs for monitoring the incidents, the root cause and detection to resolution corrective actions using standard Commercial off-the-shelf based Investigation Management tool	1	3	-	1
PC15. identify critical security threats and vulnerabilities across all layers of the IoT solution	2	2	-	1
PC16. predict and extrapolate attack trends ahead of their occurrence	2	2	-	1
PC17. carry out backups of security devices and applications in line with security policies, procedures and guidelines	1	3	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8232
NOS Name	Detect occurrences of security incidents to IoT solutions
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	6
Credits	5
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8233: Respond to detected security incidents and restore affected capabilities

Description

This unit is about responding to security incidents, preventing their expansion and ensuring timely restoration and recovery of affected capabilities

Scope

The scope covers the following :

- Drive response planning, Manage communications, Prepare preliminary reports, Drive restoration and recovery

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** plan timely response to detected security incidents
- PC2.** execute post-incident processes and procedures in line with security policies, procedures, and guidelines
- PC3.** automate responses to detected security threats and incidents
- PC4.** assign information security incidents promptly to appropriate people for investigation/action
- PC5.** track the progress of investigations into information security incidents and escalate to appropriate people where progress does not comply with standards or service level agreements (SLAs)
- PC6.** . liaise with stakeholders to gather, validate and provide information related to information security incidents, where required
- PC7.** report to law enforcement agencies, if required
- PC8.** prepare accurate preliminary reports on information security incidents using standard templates and tools
- PC9.** submit preliminary reports promptly to appropriate people for action
- PC10.** neutralize the effects of the security incident or mitigate it
- PC11.** . prevent further expansion of the security incident
- PC12.** ensure timely restoration of IoT assets and systems affected by the security incident
- PC13.** incorporate lessons learned to prevent future security incidents
- PC14.** update the organization's knowledge base promptly and accurately with information security incidents and how they were managed

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to maintaining solution security

Qualification Pack

- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies for documenting and implementing security procedures
- KU4.** who to involve while defining and executing security processes
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** the fundamentals of security and the various levels that need to be covered such as device, cloud, communications, applications, etc.
- KU7.** the routine operational procedures and tasks required to be coordinated and responded to
- KU8.** different stages of incident management such as identifying, containing, cleansing, recovering, and closing and how to perform each task respectively
- KU9.** how to restore and recover systems after a security incident
- KU10.** the common security issues and incidents that may require action and whom to report them to
- KU11.** how to obtain and validate information related to security issues
- KU12.** how to prepare and submit security reports and whom to share them with
- KU13.** how to query and write scripts
- KU14.** how to identify and refer anomalies in data
- KU15.** how to store and retrieve information
- KU16.** how to keep track of changes, procedures, and practices

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** refer anomalies to the supervisor
- GS7.** ask for clarification and advice from appropriate people
- GS8.** analyze the business impact and disseminate relevant information to others
- GS9.** apply balanced judgments to different situations
- GS10.** check the work is complete and free from errors
- GS11.** work independently and collaboratively

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. plan timely response to detected security incidents	3	5	-	2
PC2. execute post-incident processes and procedures in line with security policies, procedures, and guidelines	2	3	-	1
PC3. automate responses to detected security threats and incidents	3	5	-	2
PC4. assign information security incidents promptly to appropriate people for investigation/action	3	5	-	2
PC5. track the progress of investigations into information security incidents and escalate to appropriate people where progress does not comply with standards or service level agreements (SLAs)	1	3	-	1
PC6. . liaise with stakeholders to gather, validate and provide information related to information security incidents, where required	2	2	-	1
PC7. report to law enforcement agencies, if required	3	5	-	2
PC8. prepare accurate preliminary reports on information security incidents using standard templates and tools	1	3	-	1
PC9. submit preliminary reports promptly to appropriate people for action	1	3	-	1
PC10. neutralize the effects of the security incident or mitigate it	3	4	-	2
PC11. . prevent further expansion of the security incident	1	3	-	1
PC12. ensure timely restoration of IoT assets and systems affected by the security incident	3	5	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. incorporate lessons learned to prevent future security incidents	2	2	-	1
PC14. update the organization's knowledge base promptly and accurately with information security incidents and how they were managed	2	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8233
NOS Name	Respond to detected security incidents and restore affected capabilities
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	6
Credits	3
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8238: Create technical documents and manuals

Description

This unit is about creating various types of technical documents and manuals including online configuration manuals, user helps, solution information, installation guides, etc.

Scope

The scope covers the following :

- Plan the documentation process
- Develop drafts for the document
- Publish the document

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify the purpose and the scope of the activity for which technical documentation is to be produced
- PC2.** obtain information for the technical document from relevant sources and stakeholders
- PC3.** draft technical document ensuring that content is concise, complete, and easy to consume
- PC4.** review technical document content with relevant stakeholders and document owners
- PC5.** ensure that technical document is formatted and designed as per specifications
- PC6.** transfer technical documents to relevant stakeholders for sign-off and publishing
- PC7.** continuously review and update technical document

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to drafting and publishing technical documentation
- KU2.** organizational policies and procedures for sharing data
- KU3.** the range of standard templates and tools available and how to use them
- KU4.** the solution for which documentation is to be created
- KU5.** how to create clear and concise content
- KU6.** how to format and align document

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately

Qualification Pack

- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** refer anomalies to the supervisor
- GS7.** apply problem-solving approaches to different situations
- GS8.** analyze the business impact and disseminate relevant information to others
- GS9.** apply balanced judgments to different situations
- GS10.** check the work is complete and free from errors

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Identify the purpose and the scope of the activity for which technical documentation is to be produced	5	11	-	3
PC2. obtain information for the technical document from relevant sources and stakeholders	5	8	-	3
PC3. draft technical document ensuring that content is concise, complete, and easy to consume	5	8	-	3
PC4. review technical document content with relevant stakeholders and document owners	3	4	-	2
PC5. ensure that technical document is formatted and designed as per specifications	3	4	-	3
PC6. transfer technical documents to relevant stakeholders for sign-off and publishing	3	4	-	3
PC7. continuously review and update technical document	6	11	-	3
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8238
NOS Name	Create technical documents and manuals
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	5.5
Credits	1
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8249: Operational Technology Security

Description

Operational Technology Security

Scope

The scope covers the following :

- Operational technology security

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand OT fundamentals and the importance of security in industrial environments.
- PC2.** Elaborate the difference between IT and OT environments
- PC3.** Discuss OT network topology components
- PC4.** Familiarize with international standards and compliance requirements.
- PC5.** Learn how to implement segmentation and zone-based security and ensure OT networks are segmented from IT networks to reduce attack surface and prevent lateral movement of threats.
- PC6.** Maintain an inventory of all OT devices and systems, including legacy equipment and proprietary systems not on standard networks.
- PC7.** Evaluate the Common vulnerabilities in OT devices legacy hardware, firmware & conduct regular assessments tailored for industrial control systems, including firmware and hardware vulnerabilities unique to OT devices.
- PC8.** Implement strict controls over modifications to OT systems to prevent accidental or malicious changes that could impact safety or operations
- PC9.** Deploy OT-optimized intrusion detection systems IDS and continuous monitoring solutions capable of handling real-time data flows and specialized protocols (e.g., Modbus, DNP3, Proprietary protocols and detect and respond to OT security incidents.
- PC10.** Balance security with operational safety.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to designing, assembling, and manufacturing solution components
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting component designs and specifications
- KU4.** who to involve while developing and transferring designs
- KU5.** the range of standard templates and tools available and how to use them

Qualification Pack

- KU6.** different hardware design options and how to select the best option to meet the requirements and constraints
- KU7.** the production and manufacturing capabilities of the organization
- KU8.** the production and manufacturing capabilities of external vendor partners
- KU9.** how to ensure physical readiness and security of devices
- KU10.** various component certifications
- KU11.** production level scripting
- KU12.** techniques and tools to simulate, analyze and synthesize design options
- KU13.** types of hardware description languages & tools and how to use them, such as Verilog, VHDL, Field Programmable Gate Arrays (FPGA), DVT (Design, Validation, Test), firmware fundamentals, etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** refer anomalies to the supervisor
- GS7.** ask for clarification and advice from appropriate people
- GS8.** analyze the business impact and disseminate relevant information to others
- GS9.** provide opinions on work in a detailed and constructive way
- GS10.** apply balanced judgments to different situations
- GS11.** check the work is complete and free from errors
- GS12.** work independently and collaboratively
- GS13.** contribute to the quality of teamwork

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Understand OT fundamentals and the importance of security in industrial environments.	3	5	-	2
PC2. Elaborate the difference between IT and OT environments	3	5	-	2
PC3. Discuss OT network topology components	3	5	-	2
PC4. Familiarize with international standards and compliance requirements.	3	5	-	2
PC5. Learn how to implement segmentation and zone-based security and ensure OT networks are segmented from IT networks to reduce attack surface and prevent lateral movement of threats.	3	5	-	2
PC6. Maintain an inventory of all OT devices and systems, including legacy equipment and proprietary systems not on standard networks.	3	5	-	2
PC7. Evaluate the Common vulnerabilities in OT devices legacy hardware, firmware & conduct regular assessments tailored for industrial control systems, including firmware and hardware vulnerabilities unique to OT devices.	3	5	-	2
PC8. Implement strict controls over modifications to OT systems to prevent accidental or malicious changes that could impact safety or operations	3	5	-	2
PC9. Deploy OT-optimized intrusion detection systems IDS and continuous monitoring solutions capable of handling real-time data flows and specialized protocols (e.g., Modbus, DNP3, Proprietary protocols and detect and respond to OT security incidents.	3	5	-	2
PC10. Balance security with operational safety.	3	5	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8249
NOS Name	Operational Technology Security
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8231.Design processes to manage security threats and incidents across multiple technology layers	30	50	-	20	100	17
SSC/N8232.Detect occurrences of security incidents to IoT solutions	30	50	-	20	100	17
SSC/N8233.Respond to detected security incidents and restore affected capabilities	30	50	-	20	100	17
SSC/N8238.Create technical documents and manuals	30	50	-	20	100	17
SSC/N8249.Operational Technology Security	30	50	-	20	100	17
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	15
Total	170	280	-	100	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.